Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
C		2522428

Research on Olympic Medal Prediction and Influencing Factors Based on Random Forest

Summary

The Olympics is a major international multi-sport event held every four years, with the spirit about fairness, respect, pursuit of excellence, and fostering global unity and friendship through sports.

In problem I: We establish a model model based on **the Random Forest Regression** to predict the medal table in USA summer Olympics in 2028. We use **ARIMA** and other models to predict the features, predicting that the top three would be **the United States, China and Japan**, and visually display the model performance and prediction results.The confidence interval of the prediction results is obtained by **the boostrap method**. Compared with 2024, based on the difference in the total number of medals and the number of gold medals, **Japan**, **Australia**, and **the United Kingdom** are most likely to have an upward trend, while **China** and **France** are most likely to have a downward trend. As for the possibility of non-winning countries winning medals for the first time, we use **classification decision tree** to test the medal prediction results of the random forest regression model of non-winning countries, and doubly prove that five countries, namely **Samoa**, **Angola**, **Lebanon**, **Aruba**, and **Bangladesh**, may win medals for the first time. The probabilities of them winning at least one medal are **0.9924**, **0.9645**, **0.7368**, **0.6875**, and **0.6875** respectively.We calculate **the probability** of each country winning a medal in each event based on data from each award. The **relationship between countries and events** is viewed separately through both country and **sports** perspectives.**The country should select more advantageous events to improve the rank**.

In problem II:We newly define **Award Rate** as a variable to assess **the effect of great coaches**. Then the **Cook's Distance** is calculated and the threshold is set to **0.5 as an indicator** to screen for **outliers**, which is tested to be a **correct selection**. We then defined the **contribution of the great coach effect** based on award rate.The **five** sports most significantly influenced by great coaches are identified through the contribution values of coaches. Then, based on **the Topsis model**, the scores of each country on the need to introduce coaches in the five sports are obtained. The top three countries are obtained, and suggestions are given: **Britain**, **the United States**, and **Australia** could introduce coaches in **athletics**, **athletics**, and **aquatics**, respectively, with the quantitative values of their influence being **0.931**, **0.912**, and **0.918**.

In problem III:We improve the **gender ratio**, **number of participants**, and **number of events** in the training set for 2024 data respectively, increase the consideration of **groups of countries with a single advantage in a sport**, compare the predicted and actual medal rankings, and make recommendations for countries to improve their rankings.

Studying the medal table can help understand the distribution of medals in various sports, guiding countries to better develop their sports culture and promote the Olympic spirit equally.

Keywords: Machine Learning Data Processing, Random Forest Regression, Factor Analysis, Olympics medal, “Great Coach” Effect

Contents

1 Introduction

1.1 Background

The Olympic Games, as the world's most prestigious multi - sport event, have always been a symbol of international athletic competition and national pride. The Olympic Games always attract global attention, and the "medal table" serves as a key indicator to measure a country's athletic achievements.

In addition, the Olympics also represents historical milestones for many countries. For instance, Albania, Cabo Verde, Dominica and Saint Lucia won their countries' first Olympic medals at the Paris games, with Dominica and Saint Lucia even earning a gold medal each. However, more than 60 countries have yet to win an Olympic medal. Such a widely ranging medal distribution among countries indicates the need for in-depth analysis and understanding of the factors influencing medal counts.

The predictions of Olympic medal counts have always been of great interest. Conventionally, these predictions are often made close to the start of an upcoming Olympics, relying on information about current athletes scheduled to compete. But historical medal counts also carry valuable information that can be exploited to build more comprehensive and accurate models. This analysis not only satisfies the curiosity of sports fans but also provides valuable insights for countries to formulate their sports development strategies.

1.2 Our works

In the past, many studies have adopted two main strategies to predict the number of Olympic medals. One strategy is to take into account macro factors such as gross domestic product (GDP) and national influence as feature variables of the prediction model (Schlembach et al., 2021). The other strategy focuses on the time dimension and directly uses historical medal data to predict the medal situation of the next Olympic Games (XieXiaoPeng, 2018).

In contrast, the perspective of this study is more microscopic. Specifically, while considering historical medal data, the article uses **the athlete data** of the country's corresponding sports as the basic data unit and introduces a series of more detailed features to predict the number of medals. These features include the number of events and their changing trends, the number of athletes participating in specific sports in the country, the gender ratio of participating athletes, etc.

In terms of model selection, we use **the Random Forest Regression Model** as the core model, and follow the **"Machine Learning Data Processing"** process to complete the four steps of data processing, EDA, machine learning, and visualization to solve the actual medal number prediction problem, such as the possibility of a country that has not won an award for the first time, the manifestation of the coaching effect, etc. At the same time, in order to enhance the reliability and persuasiveness of the prediction results, we also introduced the method of sorting by approximate ideal solutions (TOPSIS), decision tree classification models, etc. When dealing with specific problems, these models are used to verify the prediction results of the random forest regression model. In addition, we have carried out a detailed analysis of the selected features and model performance to deeply explore the degree of influence of each feature on the prediction results, as well as the performance and effectiveness of the model in different scenarios.

2 The Description of the Problem

2.1 Restatement of the Problem

We are supposed to establish a model to predict the medal counts for each country, as well as evaluating the impact of different factors on the medal counts. Based on this model, we can get more information about the Olympics and provide guidance for the future Olympic Games.

2.2 Analysis of Specific Issues

2.2.1 Analysis of Problem 1

In the problem 1, we need to build a medal prediction model for each country, which can predict the number of gold medals and the total number of medals of the country. The model is evaluated and its precision is described. Based on this model, we need to provide a prediction interval for the medal table in the Los Angeles, USA summer Olympics in 2028 and describe the performance of each country compared to 2024.

We also need to improve the basic model to predict how countries that have not won a medal before are likely to perform. The model needs to be able to reflect the impact of the event's attributes on the outcome and help countries do events selection.

2.2.2 Analysis of Problem 2

In the problem 2, it is stated that the award situation may be caused by a "great coach" effect. However, we can only use the data provided in the question, i.e., the coaching effect is an uncaptured characteristic variable, so we need to find variables that can be used to assess the great coaching effect and corroborate our chosen effect with known coaching effects. At the same time, we need to measure how much the great coach effect contributes to the medals and pick countries with similar characteristics to present conclusions about the introduction of coaches.

2.2.3 Analysis of Problem 3

In the problem 3, we focuse on two key aspects. On the one hand, it explores the unique insights revealed by the model when predicting the number of Olympic medals, that is, in addition to the conventional prediction results, what hidden information or patterns about the number of Olympic medals can the model dig out; On the other hand, it explains how these insights can provide valuable references for the Olympic Committees of various countries and help them make more reasonable decisions in Olympic-related affairs.

3 Assumptions and Justification

To simplify the problem and make it convenient for us to make predictions, we make the following basic assumptions, each of which is properly justified.

- **No tied athletes**. There is one and only one gold, silver and bronze medalist in each event at every Olympic Games.
- **All participating countries have participated in the competition**. There are no countries that have never participated in the Olympics except 2028.

- **The composition of the 2028 Olympic program is exactly the same as the announced plan**. Assuming that boxing, weightlifting and modern pentathlon are not included in the program composition of the 2028 Olympic Games(LA28, n.d.).
- **The 'greatest coach' effect can be quantified**. When analyzing the "great coach" effect, it is assumed that the coach's experience, historical achievements, and coaching time can be quantified, and these quantitative indicators can effectively reflect the influence of coaches on the number of medals.

4 Model Overview

In order to thoroughly investigate the Olympic medal counts, we follow **the "Machine Learning Data Processing"** process. We establish a model based on **Random Forest Regression** to explore the relationship between various factors and the number of medals. We also evaluate the performance of the model.

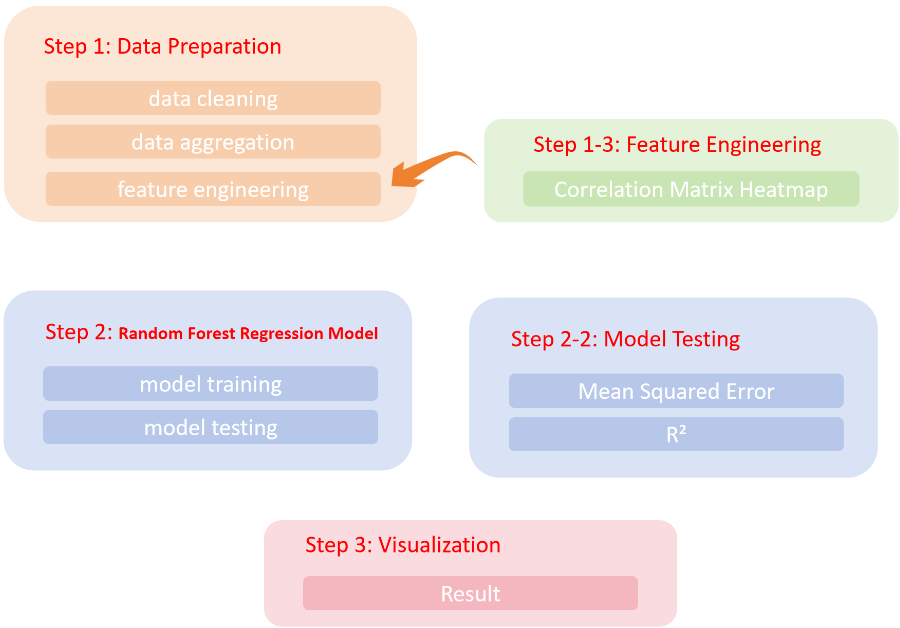

Figure 1: Model FlowChart

4.1 Data Preparation

Initially, we conduct data preprocessing. We perform data completion, deduplication, and anomaly handling on athletes, hosts, medal counts, and programs datasets.

We start from the athlete's point of view and split the Event column of each athlete after data de-emphasis. Considering that the medals of team events are unique, the data rows with the same year, country, event and award are counted and deduplicated for the medal situation. Then, we counted the total number of medals, the number of gold medals, the number of participants, the sex ratio of athletes (number of males/total number of athletes) and the percentage of newcomers (athletes who have never competed in an Olympic Games) according to the three characteristics of the Sport category, the year, and the country of the athlete in the year of the entry, and linked them to a table of the statistics of programs to obtain the number of the program of the country in that year.

The data obtained above determines whether the country was the host that year based on the year of organization and the participating cities; if so, the Host column is marked as 1, and if not, it is marked as 0.

For the same country in the same sport, the years of participation are sorted in ascending order, starting from 1896, and the number of changes in the sport for that country in the next Olympics is calculated in that order. If the data for any year is missing, it means that the country did not participate in the sport in that year, and the data is filled in with zero for the calculation.

Because athletes, coaches and training environments within the same Sport are often interconnected and share commonalities, we performed a grouped summation of award situations in athletes for the same year, using **NOC** and **Sport** as keys. We also merge the changes in the number of events within each sport from programs to explore the relationship between the number of awards and events.

In addition, since there are cases where multiple athletes in team events receive the same award, to avoid double counting the same medal in the summation, we only retained the first athlete who won a gold, silver, or bronze medal in the same event in the same year. Here, we assume that there are no cases in which multiple athletes share a medal in the same event.

Finally, we add a new column that indicates whether the country is the host in each year. The merged table is named "final_c".

4.2 Feature Engineering

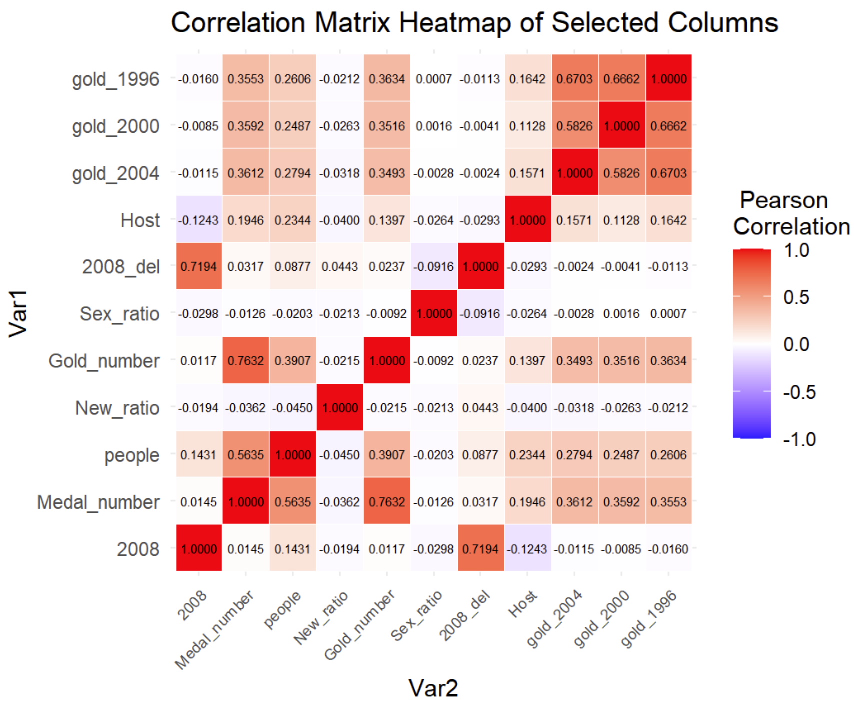

Figure 2: Corelation Matrix Heat Map

In the feature engineering step, we extract data for each Olympic Games from 1960 to 2024 for each country and Sport from the "final_c" file. We then supplement this data with the historical award-winning situations of the top three previous editions for each country and Sport as new columns. Through a **Correlation Matrix Heat map**(as shown in Figure 2), we study the correlations among various factors and ultimately selected the gender ratio, the total number of athletes participating in the competition, the number of new athletes participating, whether the country is the host, and the award-winning situations of the top three previous editions as feature values.

4.3 Random Forest Regression

Subsequently, we establish a Random Forest Regression model, using data from the two preceding editions to train the model to predict the number of medals and gold medals for the current edition. In this step, data prior to 1960 are not considered due to the irregular occurrence of the Olympic Games, which were not held every four years, and the suspension of several editions due to world wars. From 1960 to 2024, we iterate through the training and prediction of the model, obtaining the predicted and actual values for each year. The results are shown in the Figure 3.

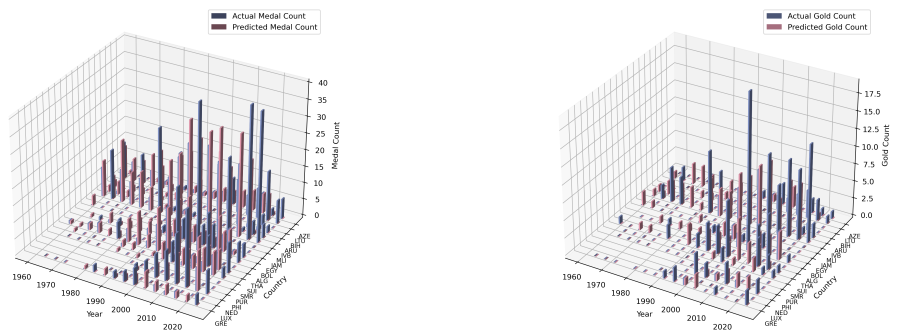

Figure 3: Total Medal and Gold Medal Number

We evaluate the model, with the following results(as shown in Figure 4):

- For the total number of medals, the average Mean Squared Error (MSE) is approximately 1.29, and the average R^2 value is about 78.1%.
- For the number of gold medals, the average Mean Squared Error (MSE) is approximately 0.31, and the average R^2 value is about 71.5%.

```
Mean Squared Error: 1.2893309147398246
Medal r2: 0.7814009181440514
Mean Squared Error: 0.31014574497415
Gold r2: 0.7150650578993729
```

Figure 4: MSE and R^2

When predicting the situation of the 2020 Tokyo Olympic Games, the prediction error is relatively large. We speculate that this is due to special factors such as the pandemic and the postponement of the Olympics. Overall, the model's prediction performance is good.

5 Problem I: Medal Counts

5.1 Medal Table Prediction for 2028

5.1.1 Prediction Medal Table

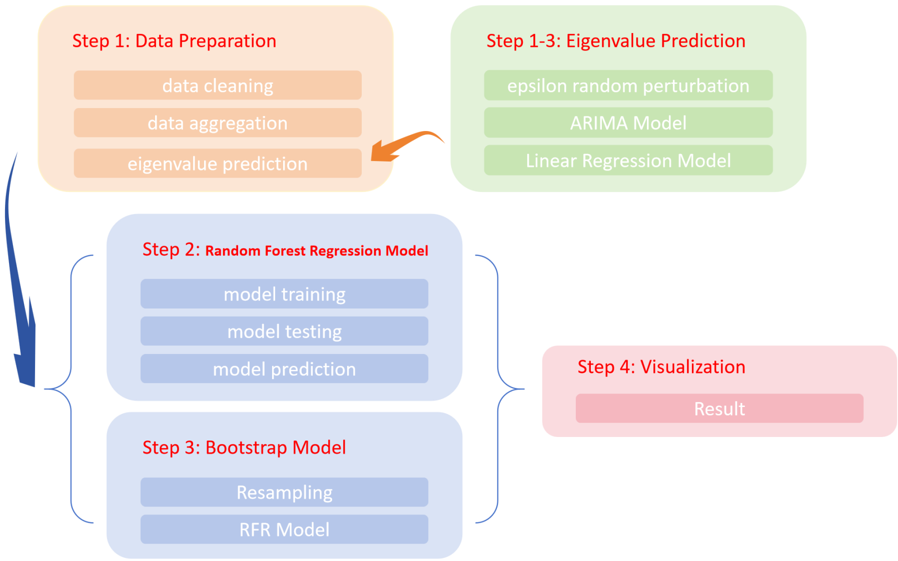

Figure 5: Prediction Model Based on Random Forest Regression

In order to predict the medal table in 2028, it is first necessary to predict the number of medals won by each country and sport in 2028. Based on the previous work, we have obtained the dataset for 2024. In order to obtain the eigenvalues for 2028, we first need to make a prediction of the features. Here we divide the data into three categories:

- For data that do not increase or decline linearly, such as gender ratio, proportion of new athletes, etc., we use **epsilon random perturbation** to predict the data for 2028 based on the addition of random perturbations to the 2024 entry data.
- For data related to time growth, such as the number of events and the increment of events, we use **the ARIMA Model** for analysis and forecasting to obtain a forecast value for 2028.
- For data with a strong linear relationship in time, such as the total number of participating athletes, we use **the Linear Regression Model** for linear regression fitting to predict the number of participating athletes in 2028 by year.

Then, the Random Forest Regression Model is trained with the data from 2024 to predict the 2028 award based on the calculated eigenvalues for 2028. The prediction results are shown in the Figure 6.

5.1.2 Prediction interval based on Boostrap method

The prediction results of the Random Forest Regression Model depend largely on the training data used. Given that different training data sets often lead to different prediction results, this study introduces **the Bootstrap method** to evaluate the stability and uncertainty of the prediction

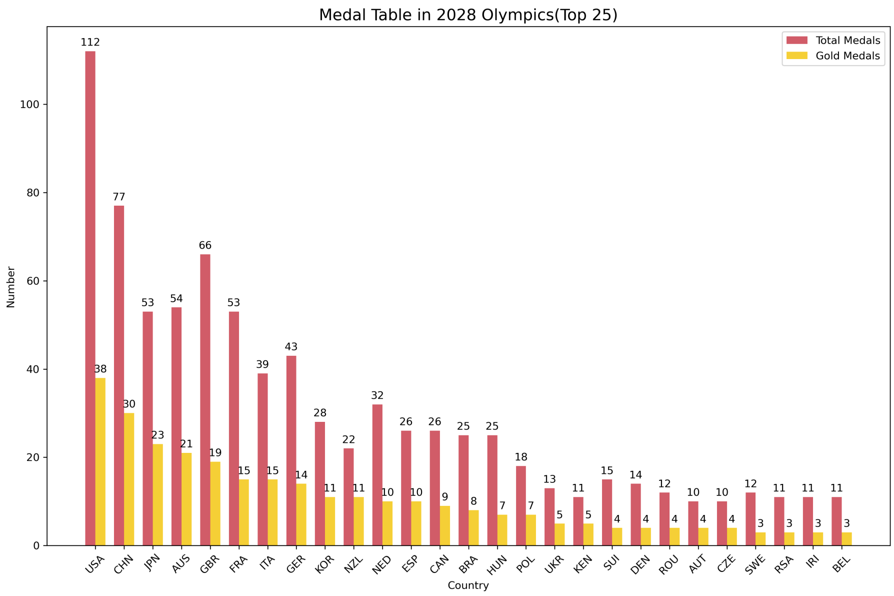

Figure 6: The Prediction Medal Table of 2028

value. This method generates multiple Bootstrap samples by resampling the original training data with replacement, and then trains the model multiple times based on these samples and obtains multiple prediction results. Based on these prediction results, we can use the percentile method to calculate the corresponding confidence interval.

Unlike traditional confidence interval calculation methods, which usually require specific assumptions about the overall distribution (such as the normal distribution assumption), the Bootstrap method is a **non-parametric** method that does not require prior assumptions about the overall distribution and is therefore more flexible and versatile.

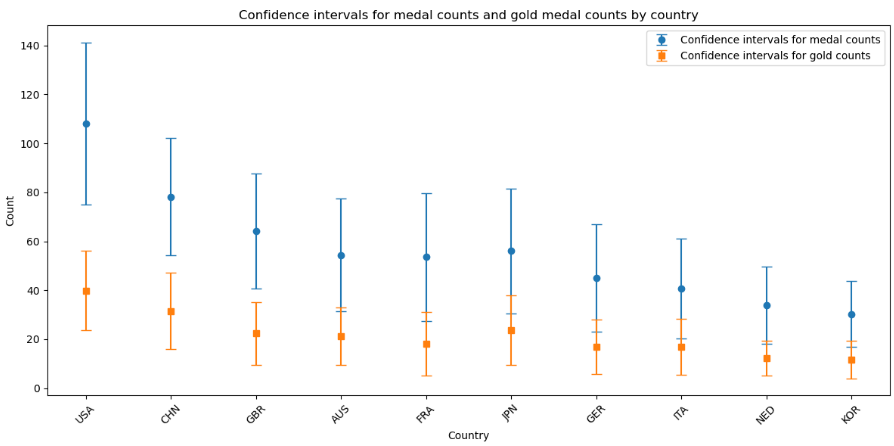

Figure 7: Predicted confidence intervals for the top ten countries in the medal table

In the specific implementation process, this study conduct 1000 Bootstrap sampling and model training. For each sample obtained by sampling, the random forest regression model is trained and used to predict the test data to obtain a series of predicted values. Subsequently, based on these 1000 predicted values, the percentile method is used to calculate **the** 95% **confidence interval**. Specifically, the 2.5th percentile and the 97.5th percentile are used as the lower and upper limits of the confidence interval, respectively.

In addition, in order to show the uncertainty of the prediction results more intuitively, this study also visualize the confidence intervals of the number of medals and gold medals of the top ten countries in the medal table to clearly show the fluctuation range of the number of medals won by different countries.

5.1.3 Comparison of medal tables in 2024 and 2028

In this study, the total number of medals and gold medals predicted by the random forest regression model for each country in 2028 are compared and analyzed with the actual data in 2024, and visualized from three dimensions: the number of medals, the number of gold medals and ranking. To ensure data integrity and comparability, the ranking of countries that did not win medals in 2024 is set to 150.

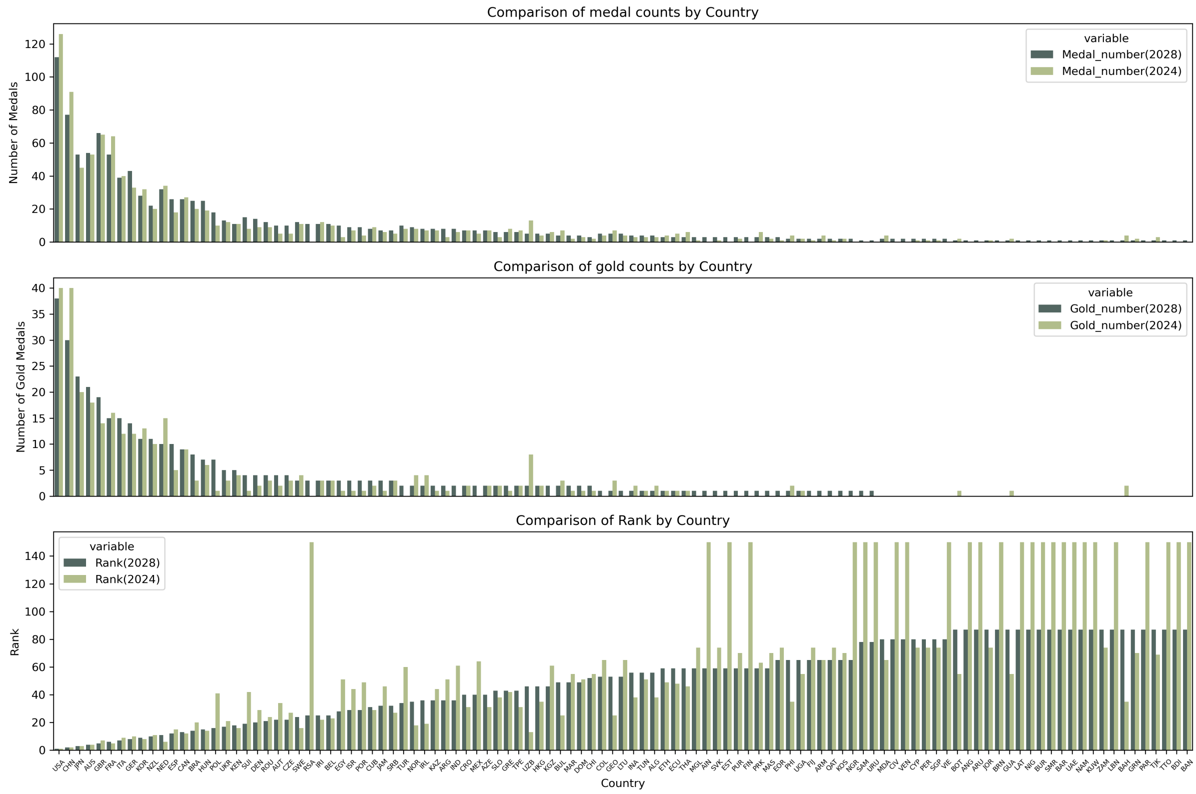

Figure 8: Visualization of the medal table in 2024 and 2028

After analysis, it is found that some countries showed an **improvement** in the comparison between the predicted data and the actual data. Among them, the countries with the **largest increase** in the total number of medals and gold medals predicted in 2028 are **Japan**, **Australia** and **the United Kingdom**. It is worth noting that these countries have relatively good international relations with the United States, the host of the 2028 Olympics, which may be positively affected by **the host effect**. Taking Japan as an example, the data shows that compared with 2024, Japan's predicted number of medals in 2028 has increased by **8** and the number of gold medals has increased by **3**. Japan continues to maintain strong competitiveness in traditional advantageous projects such as judo and table tennis; at the same time, Japan has also made significant breakthroughs in emerging Olympic projects such as rock climbing and skateboarding. This series of factors has jointly led to an upward trend in the number of medals and gold medals in Japan.

However, some countries have shown signs of **regression**. The countries with the **largest decline** in the total number of medals and gold medals predicted for 2028 are **China** and **France**. Although China still ranks second in the medal list predicted for 2028, the number of medals has decreased by **14** and the number of gold medals has decreased by **10** compared with the actual data in 2024. In-depth exploration of the reasons, **the cancellation of the weightlifting event** in the 2028 Olympics is a factor that cannot be ignored. As a traditional power in weightlifting, the absence of this event has had a great impact on the number of medals and gold medals obtained. Looking at France again, its number of medals decreased by **11** and the number of gold medals decreased by **1** in 2028. This may be related to **the host effect** enjoyed by France as the host country of the Olympic Games in 2024. By 2028, France will become an ordinary participating country, and this advantage will decrease, which in turn affects the number of medals and gold medals it wins.

5.2 First-time winning country prediction

5.2.1 Data processing

In this study, we first perform data screening on the preprocessed master table. Specifically, we extract the data subsets of countries that have never won an award and those that have won an award for the first time in the history of the Olympic Games. Then, we add a column "whether_gain" to this filtered data set as an identification variable to clearly record the awards of each country in the corresponding event. If the country wins the award in the event, "whether_gain" is assigned a value of 1; if it does not win, it is assigned a value of 0. This processed data set will serve as the basic data for subsequent model training.

At the same time, the list of countries and events that have never won an award is further filtered out from the total table. This information is accurately matched with the relevant data of the 2028 Olympics to construct a data set for the final prediction.

On the basis of completing the above data screening and sorting, in order to more comprehensively characterize the data features and improve the accuracy of subsequent analysis and prediction, this study introduced two new feature dimensions for the training data set and the prediction data set. The first is "join_count", which is used to record the cumulative number of times each country participates in a specific event; the second is "pro_existed", which is used to measure the number of times the event has been implemented in the history of the Olympic Games.

Dependent Variable	whether_gain	Whether it is the first time to win an award
Feature 1	Sex_ratio	Ratio of men and women
Feature 2	join_count	Number of times the nation have participated in the project
Feature 3	pro_existed	Number of times the project has been implemented
Feature 4	people	Number of participants
Feature 5	New_number	Number of new participants in the project
Feature 6	pro_number	Number of events included
Feature 7	pro_number_del	Increase or decrease in the number of events

Table 1: the dependent variable and features

5.2.2 classical decision tree prediction

Since whether or not to win the award for the first time is a binary classification problem, the classical decision tree model is finally chosen after comparing the performance of multiple classification models.

The principle of the classical decision tree model is to pick out the most important feature and divide the data set on this node according to this feature, with the goal of minimizing the value of Gini impurity. The calculation formula of Gini impurity is:

$$Gini(p) = 1 - \sum_{i=1}^{n} p_i^2$$

where p_i is the proportion of samples of the ith class in the data set, and n is the number of categories.

The decision tree divides the data set into smaller subsets by continuously looking for the optimal splitting condition, so that the data in each subset belongs to the same category as much as possible, thereby improving the accuracy of classification.

During the model training phase, we divided the training data into training set, validation set and test set, and adopted a five-fold cross-validation strategy to ensure the generalization ability and stability of the model.

The analysis of the confusion matrix heat map of the test set shows that the model shows a certain conservative tendency in prediction, that is, the probability of data that is actually the first award being misjudged as not winning is relatively high. Despite this deviation, in the current research background and application scenarios, this misjudgment is still within an acceptable range, and the impact on the overall research conclusions and practical applications is within a controllable range.

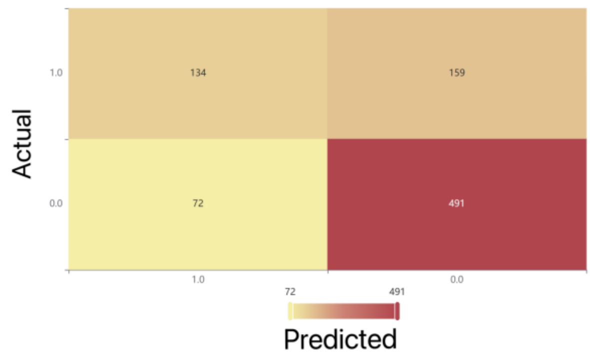

Figure 9: The confusion matrix of classification decision tree

Further analysis of the quantitative indicators shows that the accuracy, recall, precision, F1, and AUC of the test set and training set are all greater than 0.7, indicating that the model performs well and has strong predictive ability.

This model is used to predict the projects and winning probabilities of countries that did not win awards in 2028, and then the probability of countries that did not win awards winning at

	Accuracy	Recall	Precision	F1	AUC
Training set	0.767	0.767	0.773	0.75	0.834
Cross validation set	0.703	0.703	0.692	0.686	0.679
Test set	0.73	0.73	0.719	0.716	0.761

Table 2: Model evaluation indicators for classification decision trees

least one award in 2028 is calculated. The formula is:

$$P = 1 - \prod_{i=1}^{n}(1 - P_i)$$

Among them, n is the number of projects that the country is predicted to win, and P_i is the probability of failing to win the corresponding project.

In the end, there are 9 countries with a high probability of winning medals in 2028. The specific results are shown in the table.

NOC	probility
SAM	0.9923780487804879
ANG	0.9644886363636364
ESA	0.8095238095238095
NCA	0.8095238095238095
LBN	0.736842105263158
SSD	0.5454545454545455
YEM	0.75
ARU	0.6875
BAN	0.6875

Table 3: Countries that may win the award for the first time in 2028

5.2.3 Comparison of results between classical decision tree and random forest regression model

The article extracted the list of countries that did not win medals from the list of medals won in 2028 obtained by the established random forest regression model. There are five countries in total, namely **Samoa**(SAM), **Angola**(ANG), **Lebanon**(LBN), **Aruba**(ARU), and **Bangladesh**(BAN), which are all included in the results of the decision tree classification model. This directly shows that the random forest regression model correctly takes into account the countries that have not yet won medals. For emphasis, it is stated again that the probability of these five countries winning at least one medal is **0.9924**, **0.9645**, **0.7368**, **0.6875** and **0.6875** respectively.

5.3 Impact of number and types of events on national award results

As mentioned, when we prepared the train-data and validate-data, we merged the athlete and program data tables, then counted the awards won by athletes from every country, in every sport, in every year. We also calculated the number of projects per year and the incremental number of projects compared to the previous year, based on the programs data table. So our model has considered the number of sports and sports increments per project per year as characteristic variables.

5.3.1 Relationship between number of national awards and projects

To explore the relationship between the events and how many medals countries earn, we calculate the total number of awards for each program, merge it with the original data and calculate the percentage of awards per country for each program. Then we get each country's advantageous programs. In order to show the strengths of each country's program, we have aggregated by country to create a bar chart, using the United States as an example, we draw the Figure 10 .According to the plot, America is very good at Roque, Golf and Aquatics which percentage of awards is greater than 0.35.

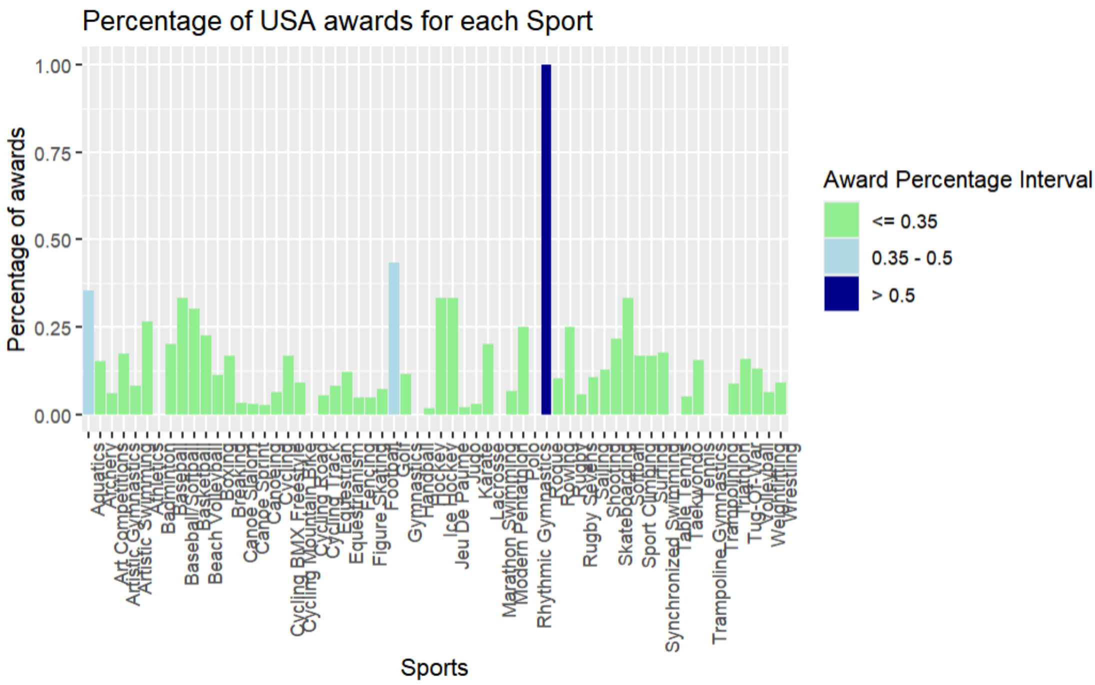

Figure 10: Percentage of USA awards for each sport

At the same time, we explore this issue from the perspective of the program, taking volleyball as an example, we draw Figure 11 to show that the three countries BRA,URS and USA are better at volleyball program.

5.3.2 Impact on results

In order to explore how country-selected programs affect the results, we first explore the importance of the number of programs selected for each country's program prediction for each year by disrupting the individual characteristic variables and comparing the MSEs for each of the validation set's number of medals and number of gold medals for importance ranking to obtain the Figure12. From the graph, it is found that the number of programs has a greater impact on the number of gold medals.

We simultaneously increase the number of an item in the validation set to 1.5 times the original number and observe the change in the medal table. We take Cycling Mountain Bike as an example, train the model with data related to 2020 and validate it with data from 2024,

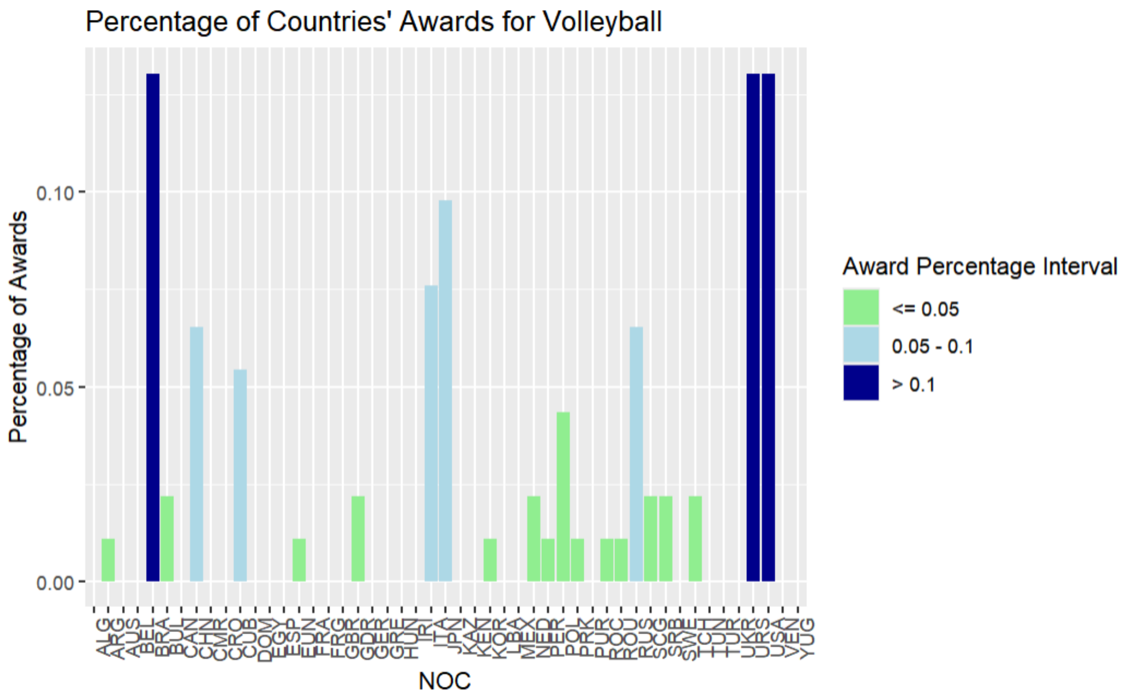

Figure 11: Percentage of countries' awards for volleyball

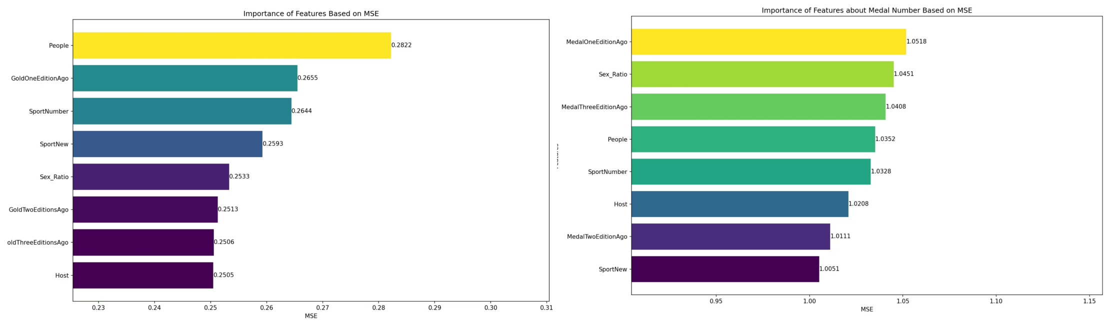

Figure 12: Importance of features about medal and Gold number

and observe the change in the medal table when the number of events is increased by the same proportion. Plotting the dumbbell graph Figure 13 reveals that countries that excel in Cycling Mountain Bike move up the rankings, suggesting that countries should choose more of their dominant sports to achieve higher rankings in the medal table(list1 is the medal table after the change in the number of events, list2 is the original medal table).

The blue line in the graph indicates the ranking of the first medal list, the red line indicates the ranking of the second medal list, the gray line connects the ranking positions of the two lists, the green dots indicate an increase in ranking, the red dots indicate a decrease in ranking, and the gray dots indicate no change in ranking.

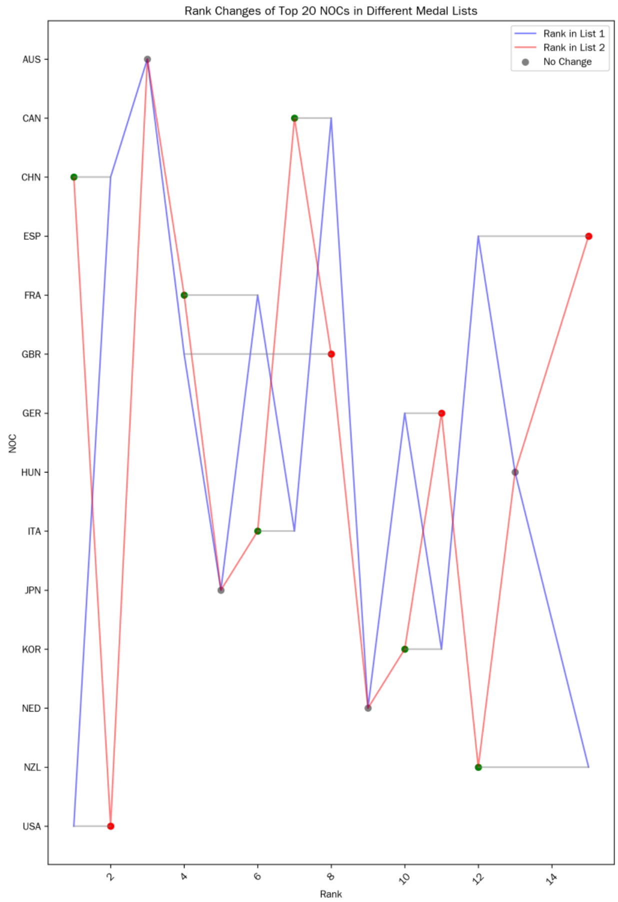

Figure 13: Change in ranking due to change in number of projects

6 Problem II: The Great Coach Effect

6.1 Selection and Testing of Indicators for Evaluating the Effects of Great Coach

In order to eliminate changes in the number of awards caused by changes in the number of entries in the program, and to magnify the gap between gold and silver,bronze medals, we define the award rate as

$$\text{Award Rate} = \frac{\text{number of gold} \times 6 + \text{number of silver} \times 2 + \text{number of bronze} \times 2}{\text{entry}}$$

to perform **outlier detection**.

Grouped by **sport and country**, counted the award rate of that country's sport in the records of previous Olympic Games, and constructed a linear model to fit the relationship between the year and the award rate.Then we calculate the Cook's Distance for each data point in each data set

$$D_i = \frac{e_i^2}{p\hat{\sigma}^2}\left[\frac{h_{ii}}{(1-h_{ii})^2}\right]$$

D_i is Cook's Distance, $e_i$represents the residual of the i - th observation, p denotes the number of explanatory variables (including the intercept term) in the regression model, $\hat{\sigma}^2$ is an estimator of the variance σ^2 of the error term, also known as the estimator of the Mean Squared Error (MSE), h_{ii} is the i - th diagonal element of the hat - matrix $H = X(X^TX)^{-1}X^T$.

Assuming a threshold of 0.5, when the Cook's Distance is greater than or equal to 0.5, we consider this data point to be an outlier, i.e., the award rate has changed significantly.

According to Olympics.com, n.d. and USA Gymnastics, n.d., we plot the changes in the award rate and Cook's Distance for all Olympic Games for volleyball in the United States and China, and the changes in the award rate and cook's distance for gymnastics in Rome and the United States, respectively.

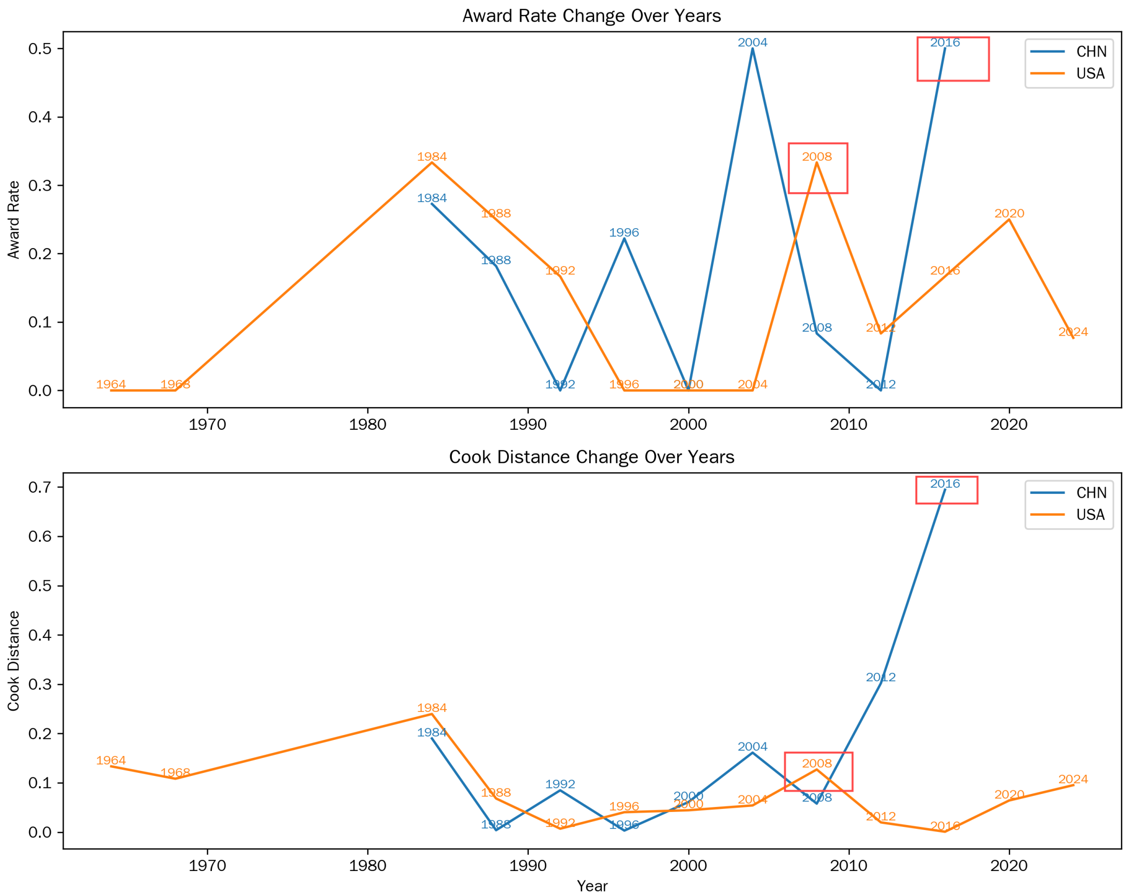

Figure 14: Volleyball's award rate changes in China and the U.S.

Meanwhile, according to the question and the reference material, comparing the line graphs, we can find that the data boxed in red are the award rates of the Olympic countries where the two coaches have made changes of nationality, which were all detected by Cook's Distance.The indicator and the test method can therefore be considered correct.

6.2 The contribution of the great coaching effect

We define the contribution of great coaches to the number of medals as the

$$\text{Contribution} = \frac{\text{true award rate} - \text{award rate predicted by last edition}}{\text{true award rate} - \text{true award rate in last edition}}$$

we then select data from the 2012-2024 Olympics to calculate the contribution of great coaches to gold medals.We obtain the contribution about country and sport in Figure16

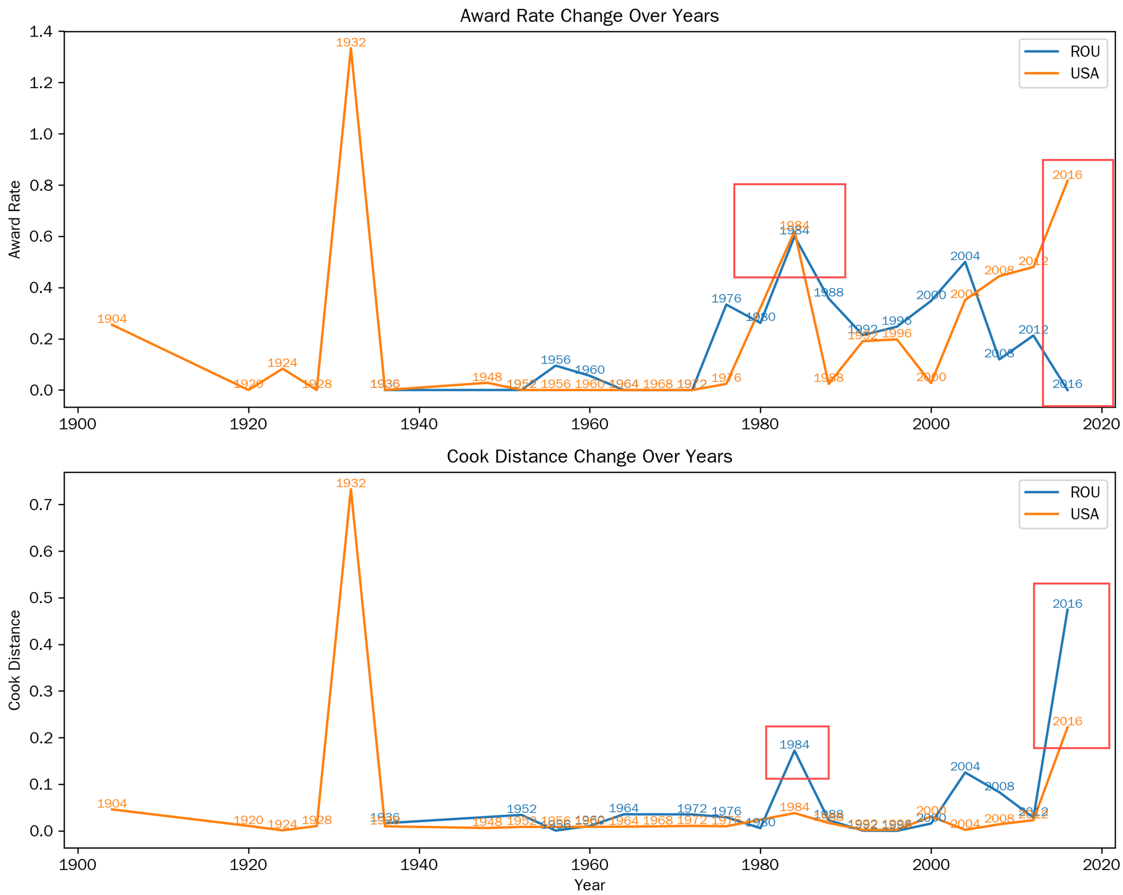

Figure 15: Gymnastics's award rate changes in Romania and the U.S.

6.3 Countries Selection by Topsis

6.3.1 Data selection

Based on the previous analysis, we have obtained the coach's contribution ratio. In order to accurately locate the impact of "great coaches", that is, to explore the ability of coaching effect to positively promote winning, this study defines the effect produced when the coach contribution rate is in the interval $[0.8, 1]$ as the great coach effect. Based on this standard, we screen out countries and sports that shows the great coach effect.

Subsequently, we conduct an aggregate analysis based on the project, calculate the average contribution ratio of coaches in each project, and count the number of times the great coach effect appeared. Taking these two characteristics into consideration, we identify the five projects that were most significantly affected by great coaches, namely: **Aquatics**, **Shooting**, **Taekwondo**, **Athletics**, and **Tudo**.

For further in-depth research, we screen out data containing the above five projects from the 2024 data set for subsequent analysis.

6.3.2 Introduction demand scoring based on Topsis model

Given that the data is known and there are significant differences between the evaluation indicators, this study uses the **TOPSIS model** to objectively score the needs of countries to introduce great coaches.

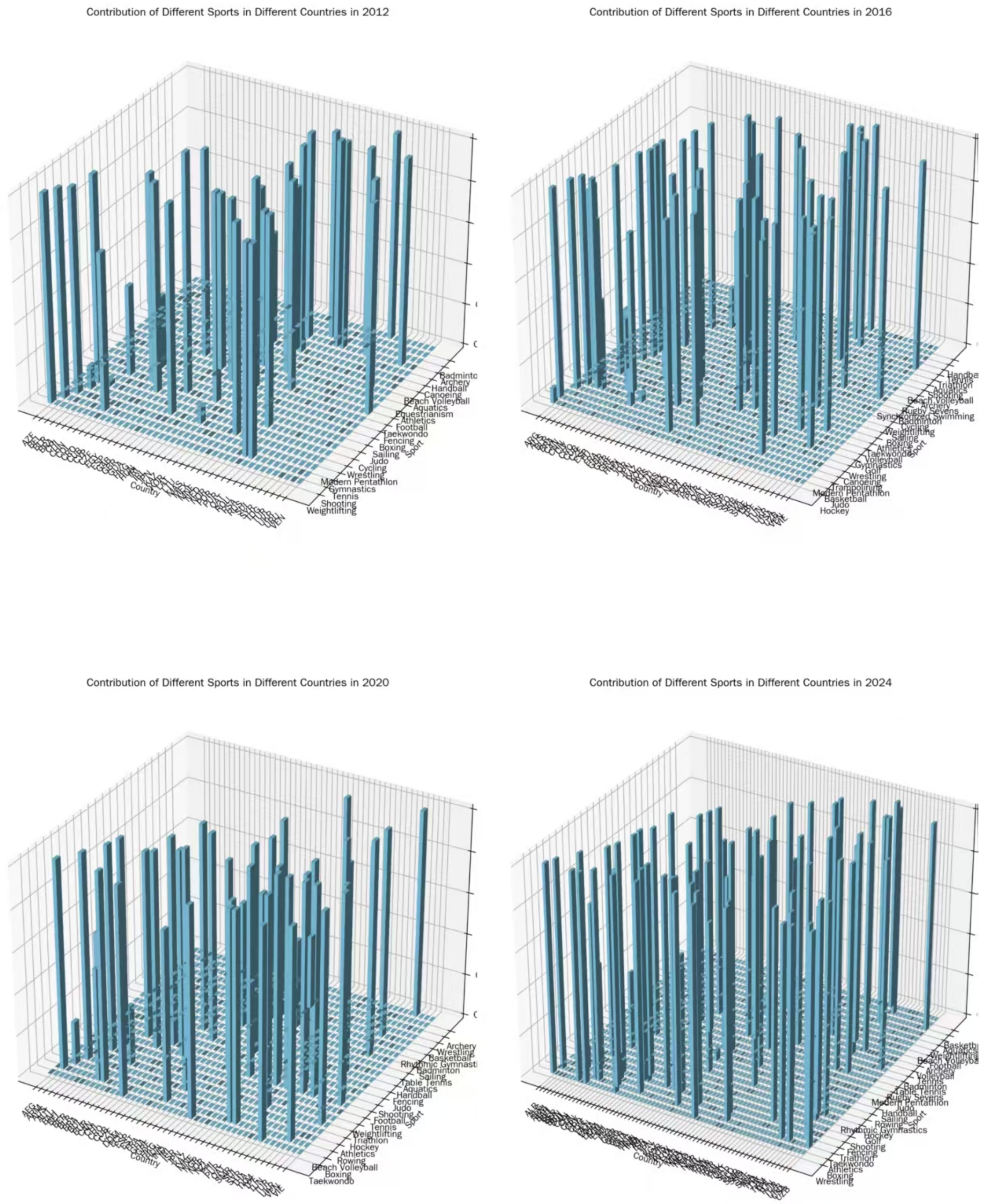

Figure 16: Estimate contribution of the great coaching effect from 2012 to 2024

In terms of data processing, countries(NOC) and sports (Sport) are selected as index items. At the same time, the nature and meaning of the following indicators are clarified: the number of participants in the project, the number of newcomers in the project, and the gender ratio of the project are set as positive indicators. This means that the larger the value of these characteristics, the more urgent the country's need to introduce great coaches in the corresponding sports. The logic is that a larger number of participants and newcomers may imply a broader talent base, but may lack effective guidance; and a reasonable gender ratio also requires professional coaches to conduct scientific guidance and training to promote the balanced development of the project. On the other hand, the total number of medals, the total number of gold medals, and whether it is the host are determined as negative indicators. That is, the smaller the value of these characteristics, the higher the country's need to introduce great coaches in the sport. A small number of medals and gold medals may indicate that the country's competitive level in this event needs to be improved, and the country needs the guidance of professional coaches to improve its performance; if it is not the host country, it may be relatively lacking in resources and experience, and it needs great coaches to bring advanced concepts and training methods.

The study uses **the Entropy Weight Method** for objective weighting, and the specific process is as follows:

1. The data need to be standardized. For the evaluation objects n and the evaluation indicators m, the original data matrix is$X = (x_{ij})_{n\times m}$, where $i = 1, 2, \cdots, n; j = 1, 2, \cdots, m.$

   The formula of positive indicator is:

   $$y_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}$$

   ,the formula of negative indicator is:

   $$y_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)}$$

   ,where $\max(x_j)$ and $\min(x_j)$ are the maximum and minimum values of the jth indicator, respectively.

2. Calculate the weight of the standardized data under each indicator p_{ij}, the formula is

   $$p_{ij} = \frac{y_{ij}}{\sum_{i=1}^{n} y_{ij}}$$

   ,where $i = 1, 2, \cdots, n; j = 1, 2, \cdots, m.$

3. Calculate the information entropy e_j of the j-th indicator. The formula is:

   $$e_j = -k\sum_{i=1}^{n} p_{ij}\ln(p_{ij}),$$

   where $k = \frac{1}{\ln(n)}$. When $p_{ij} = 0$, define $p_{ij}\ln(p_{ij}) = 0$.

The specific weights of the evaluation indicators are given in the table.

Item	Information Entropy Value e	Information Utility Value d	Weight(%)
People	0.789	0.211	43.085
New_number	0.789	0.211	43.085
Sex_ratio	0.959	0.041	8.288
Medal_number	0.999	0.001	0.224
Gold_number	0.999	0.001	0.167
Host	0.975	0.025	5.152

Table 4: Weight calculation results of entropy weight method

After clarifying the weights of each indicator, this study calculated the optimal matrix vector and the worst matrix vector by writing code. Then, for each piece of data, the distance D+ between the evaluation object and the positive ideal solution and the distance D- between the evaluation object and the negative ideal solution were calculated. These two distance values (both measured by Euclidean distance) represent the degree of proximity between the evaluation object and the optimal solution and the worst solution, respectively. Based on the above calculation results, **the comprehensive score index** is further derived. Finally, this paper visualizes the top 10 country-project combinations in terms of comprehensive scores to intuitively show their performance.

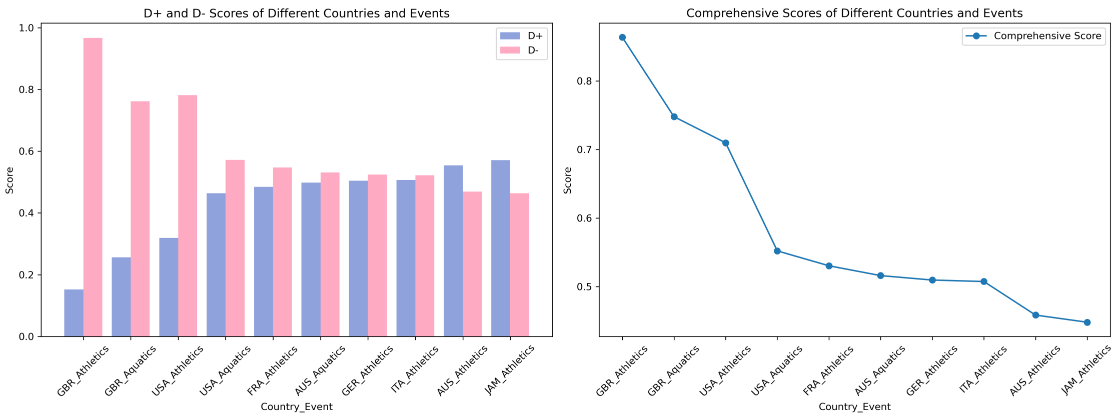

Figure 17: Top 10 Countries by Overall Score - Portfolio

6.3.3 Conclusions on investment programs and coaching impact

Based on the analysis results of the comprehensive score index and the coach effect contribution ratio, this paper puts forward the following targeted suggestions. **The United Kingdom (GBR)** can introduce excellent coaches in the **athletics** project, and the corresponding coach effect contribution ratio is **0.931**. **The United States (USA)** can also consider introducing outstanding coaches in the **athletics** project, and the corresponding coach effect contribution ratio is **0.912**, so as to fully tap the positive impact of the coach effect and promote the improvement of project performance. **Australia** can focus on **aquatics** and introduce high-level coaches. The corresponding coach effect contribution ratio of this project is **0.918**, and enhance the competitiveness of the project by optimizing the allocation of coach resources.

7 Problem III: The Insights and Suggestions

In this study, we set the 2024 dataset as the test set and use the 2020 dataset to train the random forest model, using this training as an example to illustrate our model insights and derived recommendations.

To achieve accurate predictions of the medal table, we build a prediction model by adjusting three key indicators: the proportion of male athletes in the test set, the number of people participating in the Olympics, and the number of national registrations. Subsequently, the **dumbbell plots** of comparing the predicted medal table (list1) and the actual 2024 medal table (list2) data were drawn, and the impact of the above three indicators on the prediction results was deeply analyzed, and strategic recommendations were provided to the National Olympic Committee to improve the medal table ranking.

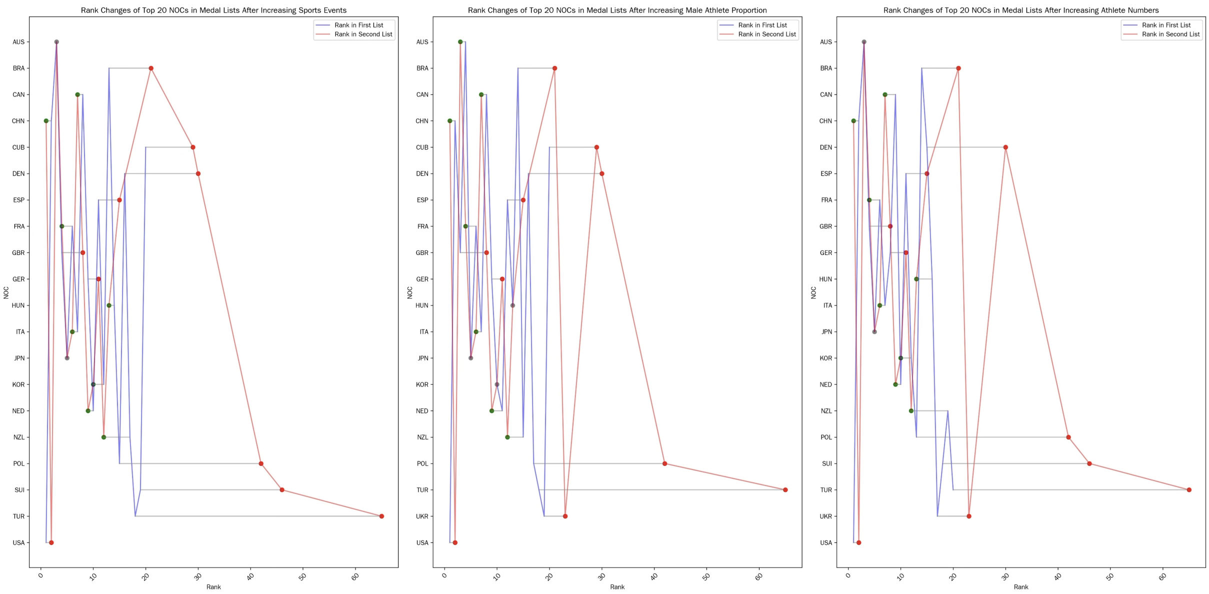

Figure 18: Dumbbell plot for comparison

Taking **USA** as an example, the United States can increase the number of gold medals and medal ranking by **increasing the proportion of male athletes**, **increasing the number of Olympic participants**, and **increasing the number of participating events**. For example, **China** can increase the number of gold medals and medal ranking by **reducing the proportion of male athletes**, **reducing the number of Olympic participants**, and **reducing the number of participating events**.

At the same time, this study further analyzes the performance of the winning countries. Specifically, the total number of medals and the number of gold medals won in each event are calculated as the proportion of the total number of medals and the number of gold medals won by the country in the current Olympic Games. The study stipulates that if the total number of medals of a country is greater than 3, and one of the proportion of the total number of medals and the proportion of the number of gold medals exceeds 0.5, the country is judged to be **a single advantage country**. The study found that when the number of sub-items of the project changes, the ranking changes of single advantage countries such as **Cuba (CUB)** and **Turkey (TUR)** are more significant than those of China and the United States, which have comprehensive and balanced development. This phenomenon shows that the ranking of countries with single-item advantages is greatly affected by the number of sub-items of the project, and their ranking

stability is relatively poor.

In the analysis of the medal table of the 2024 Olympics, we also found that countries with a single advantage accounted for 38% of the medal table, however, none of these countries entered the top 20 of the medal ranking. This result fully demonstrates that for countries with a single advantage such as **Uzbekistan (UZB)** and **Algeria (ALG)**, if they want to become sports powers in the long run, it is crucial to **actively expand and develop diversified sports** and pursue a more balanced sports development pattern.

8 Sensitivity and Robustness Analysis

8.1 Sensitivity analysis

Sensitivity analysis is to assess whether the already trained model can get accurate results when some of the test data are inaccurate. If the R-squared change is small and maintains a large level, it means that the trained model can still predict the results well, that is to say, the model is generalizable.

According to the aforementioned, we choose 2020 data for model training and 2024 data for testing, randomly disrupt the features of a column for prediction, and then count the MSE of the number of medals and the number of gold medals and sort them. The higher the value of MSE ,the more important the features are for model training. We get the Figure 12.

Since the award data of previous years is a known result, we assume that the medal and gold medal data of previous years are accurate and there will be no statistical inaccuracy. Because whether it is the host or not is a categorical variable, so we discuss it separately. We choose 2020 data for model training and 2024 data for testing, pick a column of 2024 data for 0 to 0.9 multiplicity change, other values remain unchanged, and statistically the R-square of the number of medals and gold medals, and plot the Figure19.

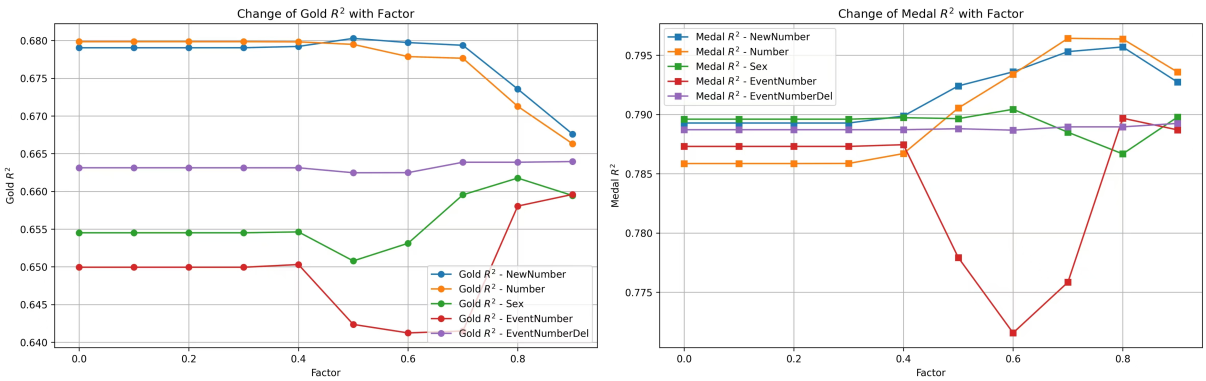

Figure 19: Change of R squared with factors

From the Figure19 , we find that even if we change any one of the features, the variation interval of R-square is very small, which indicates that the model we trained is very generalizable. From Figure 12, we find that the MSE changes less when disrupting Host, suggesting that our model can also anticipate the results better when there is a problem with the statistics of the host data, indicating that **our model is very generalizable**.

8.2 Robustness Analysis

Add noise to the training data to judge the robustness of the model. Specifically, adjust the intensity of the noise "`noise_level`" added to the training data . When "noise_level" is 0.0, it means no noise is added, that is, the original training data is used for model training; when "noise_level" is equal to 0.05, it means adding normally distributed noise with a standard deviation of 0.05 to the training data. We will visualize the goodness of fit R^2 and the mean square error (MSE) of the prediction error after adding noise of different intensities.

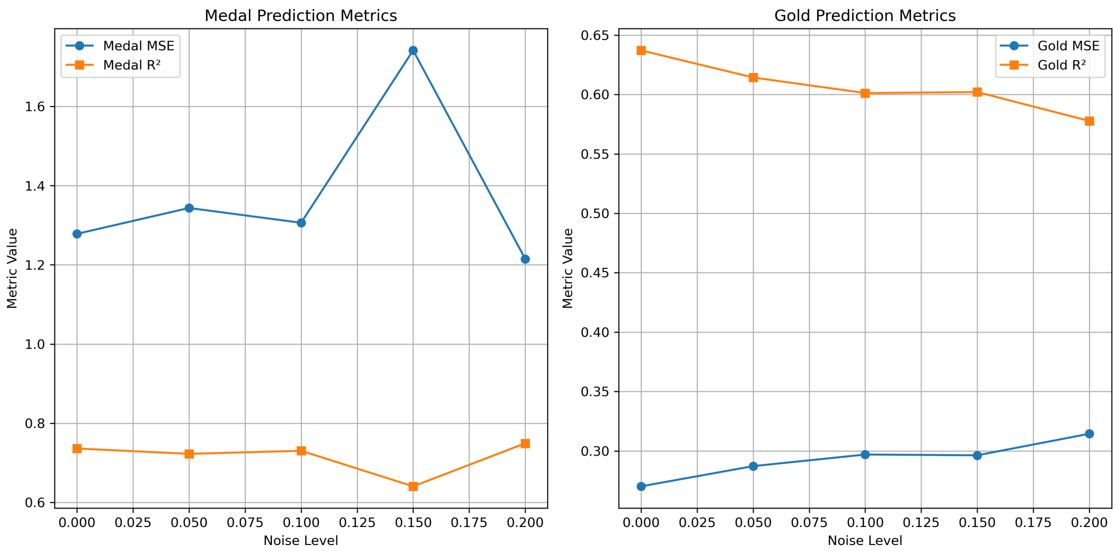

Figure 20: MSE and R^2 at different noise levels

For the prediction of the number of medals, MSE fluctuates greatly under different noise levels. When the noise level is 0.1 and 0.2, the model adapts well to some noise interference, which may mean that the model readjusts the parameters in some way under higher noise levels, so that the prediction error is reduced again; R^2 is relatively stable under the noise level, which can show that the random forest regression model has good robustness for the prediction of the total number of medals.

For the prediction of the number of gold medals, as the noise level increases from 0.0 to 0.2, the MSE shows an overall upward trend, and the prediction error of the model continues to increase, but the MSE growth rate is within 0.1, which is considered acceptable. At the same time, as the noise level increases, R^2 shows an overall downward trend, indicating that the model's ability to fit the number of gold medals gradually weakens with the increase of noise, but the decline is around 0.02, which is considered acceptable. Therefore, for the prediction of the number of gold medals, the robustness of the random forest regression model is different from the performance of the prediction of the number of medals, but it still shows that it has a certain degree of robustness.

9 Evaluation and Promotion of Model

9.1 Strength and Weakness

9.1.1 Strength

- The features of the model come from a large number of data bases and the selection of features has a calculation basis, and the establishment of the model has a reliable mathematical basis and high reliability.
- According to the Mean Squared Error and R^2 in the evaluation part of the model, the model performed well and the conclusions were reliable.
- The Random Forest Regression Model continuously adds features, processes higher-dimensional data, and can efficiently process a large amount of data, without requiring a lot of parameter adjustments to achieve ideal results, and the model is highly scalable.

9.1.2 Weakness

- To predict future medal table, it is necessary to predict the features first. The uncertainty error is large. In the sensitivity test, the model is more sensitive to the event choice of the country in the current year, and the error of event number prediction has a great impact on the model results.
- The desicion and fitting steps inside the model are black-box and less interpretable.

9.2 Promotion

In the future work, considering that athletes' participation experience has a great impact on the probability of individual awards, and the national situation has a strong correlation with the athletes' training environment. We can add the number of athletes participating in the Olympics and the national GDP as supplementary features, which can increase the accuracy of the model's predictions.

For the features with high model sensitivity, we need to obtain a more accurate data base as much as possible when predicting, and try more prediction models to obtain more accurate eigenvalue data.

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Report on Use of AI

1. Cici (Jan 25, 2025 version)

   Query1: Read csv, including Year NOC Medalnumber Gold number Sport premedal pregold a few columns. According to the year, the NOC will sum Medalnumber Goldnumber premedal pregold four columns separately and restore them.

   Output: You can use Python's pandas library to group and sum columns Medalnumber, Goldnumber, premedal, and pregold by Year and NOC, and re-store the results.

2. Kimi (Jan 5, 2023 version, Ernie 4.0)

   Query1: Help me gather some reliable news documents to tell me when Lang Ping was a volleyball coach in which country, how long his tenure has changed, and tell me where the news is based on

   Output: Coaching in Italy Modina FC: In 1989, Lang Ping became the head coach of the Italian club Modina women's volleyball team and led the team to win the Italian Cup. Modena Women's Volleyball Team: In 1999, Lang Ping became the head coach of the Modena women's volleyball team in Italy, leading the team to win the Italian Women's Volleyball League in 2000, the UEFA Women's Volleyball Champions League in 2001, and the Italian League and Cup in 2002. Novara Club: At the beginning of the 2002-2003 season, Lang Ping served as the head coach of Novara Club Italy, leading the Novara women's volleyball team to win the Italian Super Cup and the 2004 Italian League championship. Coaching in the United States University of New Mexico Women's Volleyball Team: In 1991, Lang Ping became the head coach of the University of New Mexico women's volleyball team and led the team to the Eastern Conference Women's Volleyball Championship. U.S. Women's Volleyball Team: In February 2005, Lang Ping became the head coach of the U.S. women's volleyball team. In 2007, she led the U.S. women's volleyball team to win the third place in the World Cup and get a ticket to the Beijing Olympics. In the 2008 Beijing Olympics, Lang Ping led the American women's volleyball team to win the silver medal again after 24 years. Coaching in Turkey In the 2008-2009 season, Lang Ping coached the Ankara Telecom women's volleyball team in Turkey to reach the top four of the league. The above information is mainly from Baidu Encyclopedia.